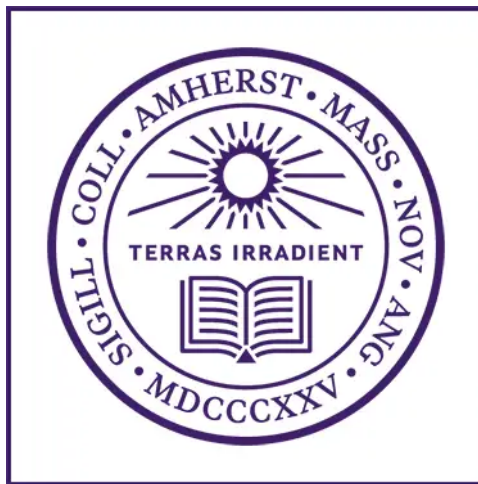


Economic Effects of Splitting Cities When Redistricting: Effects of Gerrymandering Beyond Politics

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Abstract

This thesis tests whether splitting a city across multiple congressional districts changes its fiscal receipts. Prior literature establishes that greater political representation increases fiscal outcomes at the county level, but no prior work has examined whether purely geographic fragmentation, splitting an administratively unified city across districts without changing its total representation, impacts fiscal receipts. When a city straddles two districts, each representative bears the full cost of advocacy while capturing only a fraction of the electoral return, predicting lower fiscal transfers, yet the city also benefits from two Congressional votes, predicting higher fiscal transfers. These hypotheses are tested using a national panel of municipalities spanning 1987 to 2022, drawing on seven waves of Census of Governments financial data linked to congressional district boundaries, 1990 demographic controls, Cook Partisan Voting Index scores, and nearly 100 data sources. This analysis required spatial computation and consolidation of non-standard data across state and local agencies that only became practical with new software systems.

The empirical strategy is a two-way fixed-effects event study that exploits decennial redistricting as an exogenous source of variation in municipal split status, comparing localities that transition between unified and split configurations against localities whose representational status remains stable. I find that split municipalities receive between 11% and 19% fewer federal intergovernmental transfers per capita than their unified counterparts across five Census of Governments waves spanning 1987 to 2017. Local tax revenue is unaffected, consistent with the mechanism operating through the federal representation channel. Interaction effects were observed with respect to city size and partisan alignment. The effect is concentrated among small and medium municipalities, where representatives serving split cities also represent multiple other towns, intensifying the collective action problem. Partisan alignment amplifies the penalty: fiscally unified municipalities whose representatives are co-partisan with the governor and president receive fewer transfers, consistent with taken-for-granted dynamics in the distributive politics literature. This dataset may serve as a basis for further research into the broader economic effects of redistricting and its political implications.

JEL Classification: H77, D72, R50

Keywords: Congressional redistricting, gerrymandering, municipal finance, fiscal federalism, federal intergovernmental transfers, distributive politics, government fragmentation, collective action, partisan alignment, political geography, place-based policy

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I. Introduction

Congressional redistricting is typically understood as a political process. It helps to determine which party controls a seat, which communities share a representative, and how electoral power is distributed across the country. This paper argues that redistricting also has direct economic implications. Where district lines fall relative to city boundaries impacts the fiscal resources those cities receive from the federal government. Cities whose boundaries fall entirely within a single congressional district receive approximately \$3.77 more in federal intergovernmental transfers per capita each year than cities whose boundaries are split across multiple districts, among municipalities that receive any federal transfers. Cumulated over a decade, that gap amounts to roughly \$37.65 per resident. For a mid-sized American city of 100,000 people, this implies a difference of approximately \$3.8 million in federal resources over a single redistricting cycle, determined entirely by where a line on a map is drawn.

This paper originates from a practical policy question. Leading up to the 2020 redistricting cycle, debates over the configuration of Massachusetts congressional districts raised a question that would prove surprisingly difficult to answer: should the city of Fall River be split across two congressional districts, as it had been since 2010, or should it be made whole under a single representative? The question is not merely representational; it carries significant economic weight. Fall River is a historically working-class city of roughly 90,000 residents, one of the most economically distressed municipalities in Massachusetts. Its congressional configuration directly shapes how effectively its government can advocate for federal resources, coordinate infrastructure investment, and navigate the grant programs through which district boundaries translate into fiscal outcomes. This question has personal meaning for me: in 2021, I interned for

Representative Jake Auchincloss, whose congressional district includes Fall River, and had the opportunity to observe the many ways a congressional office responded to residents' needs.

Strikingly, similar deliberations take place across the country every redistricting cycle, but lack empirical economic evidence. In theory, the fiscal impact is ambiguous: a split city benefits from having two representatives, each able to advocate for its interests on separate committees and across different legislative coalitions. Conversely, the city could be disadvantaged because two representatives for a single city are each less incentivized to advocate due to a collective action problem, resulting in less total advocacy. Separately from fiscal transfers, the political fragmentation imposed by splitting may make the ordinary business of municipal governance harder: more meetings, more offices to navigate, more competing priorities, and a weaker local business environment. This thesis aims to answer these questions empirically, using federal fiscal and political data across American cities and over three decades of redistricting cycles.

To resolve this ambiguity empirically, the paper constructs a national panel dataset of municipal governments spanning 1992 to 2022, drawing on nearly 100 data sources across all 48 contiguous states. Each source required separate data acquisition, format conversion, geographic projection, and crosswalking to consistently place identifiers across six Census of Governments waves. The result is the first dataset capable of tracking the split status of American cities across multiple redistricting cycles and linking that status to detailed fiscal outcomes at the municipal level.

The paper estimates the effect of congressional district fragmentation on two outcomes: federal intergovernmental transfers per capita and local tax revenue per capita. The primary

empirical strategy is a two-way fixed-effects event study regression that exploits the decennial redistricting cycle as a source of plausibly exogenous variation in cities' split status, comparing cities that transition from unified to split, or from split to unified, against cities whose representational status remains unchanged across the same period. Outcomes are expressed in logs to facilitate interpretation as percentage effects, and the analysis focuses primarily on the 1990, 2000, and 2010 redistricting cycles, as pre-period fiscal data are unavailable for 1980, and the effects of the 2020 cycle have not yet fully materialized.

Fall River itself illustrates the design. Unified through 2010, split across two congressional districts from 2012 through 2020, and unified again following the 2020 redistricting, Fall River experienced both transitions within the study period. Its fiscal trajectory across those transitions, benchmarked against similar cities whose representational status did not change, is precisely the kind of within-city variation the event study design exploits. The results suggest that the transition to split status is associated with an estimated cumulative reduction of approximately \$3.4 million in federal transfers over the split period, while the reunification under a single representative following the 2020 redistricting is associated with a recovery of similar magnitude.

This paper makes three contributions. First, it constructs a new national municipal panel dataset of high scope and resolution, integrating fiscal, geographic, demographic, and political data across nearly 100 sources and three decades. This dataset is designed to be replicable and extensible, and represents a contribution to the research infrastructure of municipal finance and political economy independent of the paper's empirical findings. Second, it provides the first evidence that representational fragmentation, the splitting of an administratively unified city

across multiple congressional districts, without any change in local government structure, has measurable effects on city-level fiscal and economic outcomes. This distinguishes the paper's contribution from the existing literature on government fragmentation, which focuses on formal administrative splits, and from the partisan alignment literature, which focuses on the party of the representative rather than the district's geographic configuration. Third, the paper synthesizes previously separate literatures on fiscal federalism, the political economy of distributive spending, the economics of government fragmentation, and the political geography of redistricting, into an integrated empirical framework at the municipal level.

The findings have direct implications for redistricting policy. In the United States, redistricting criteria vary considerably across states: some explicitly require that political subdivisions be kept whole where possible, while others permit, or even routinely employ, city splitting to satisfy population-equality requirements or partisan objectives. The evidence in this paper suggests that splitting cities imposes measurable economic costs on their residents, costs that have not previously been quantifiable and that have therefore been absent from redistricting deliberations. To the extent that those deliberations can be improved by better evidence, this paper provides a basis for treating municipal integrity as an economically meaningful criterion, rather than merely a representational or aesthetic one.

II. Related Literature

This paper sits at the intersection of four bodies of literature: fiscal federalism and the relationship between representation and public expenditure; the political economy of distributive spending; the economics of government fragmentation; and the political geography of

redistricting. I found no prior work on the US political economy that tested these four literatures together at the national level with municipal granularity, nor work that linked the geographic configuration of congressional districts to city-level fiscal outcomes. Prior research has achieved one or two of the three dimensions that define this paper's contribution: geographic precision at the municipal level, national scale, and a panel spanning multiple redistricting cycles, but none has combined all three. The absence of prior work on this question reflects the practical barriers to studying it: until recently, the geographic data and computational tools required to construct a national municipal-district crosswalk were not publicly available in usable form.

The closest antecedent to this paper in the existing literature is the work on political representation and fiscal outcomes, which establishes the empirical foundation on which this analysis builds. Ansolabehere, Gerber, and Snyder exploit the court-ordered reapportionment reforms of the 1960s, stemming from the Supreme Court's one-person-one-vote ruling, to show that counties that lost representation in the state legislatures subsequently received lower state transfers per capita, while formerly underrepresented counties gained funds, empirically showing that when a jurisdiction's political weight is diluted, the public resources that follow representation fall with it.¹ The international evidence reinforces and extends this finding. Bhavnani² uses India's 2008 electoral delimitation as a quasi-experiment, showing that areas gaining legislative seats experienced faster economic growth than those losing representation, while Imai³ reaches parallel conclusions in the Japanese context, finding that shifts in political

¹Ansolabehere, Stephen, Alan Gerber, and James M. Snyder. "Equal Votes, Equal Money: Court-Ordered Redistricting and Public Expenditures in the American States." *American Political Science Review* 96, no. 4 (2002): 767-777. <https://doi.org/10.1017/S0003055402000448>

²Bhavnani, Rikhil R. "The Effects of Malapportionment on Economic Development." *PLoS ONE* 16, no. 12 (2021): e0259150 <https://doi.org/10.1371/journal.pone.0259150>

³Imai, Masami. "Local Economic Impacts of Legislative Malapportionment." *Journal of Urban Economics* 131 (2022): 103475 <https://doi.org/10.1016/j.jue.2022.103475>

weight translated into measurable differences in national transfers and local economic indicators. Together, these studies establish a robust cross-national pattern in which representation shapes resource allocation and boundary changes that alter representational weight produce measurable fiscal consequences. What none of them addresses is whether purely geographic fragmentation, splitting a city across districts without any change in the per-capita number of representatives assigned at the state level, produces the same effect.

A second body of literature examines how political representation and partisan alignment shape the distribution of federal spending across districts and localities. Berry, Burden & Howell⁴ establish that the president systematically influences the geographic allocation of federal outlays, directing more spending toward districts represented by co-partisans, a finding that situates federal transfers as objects of political competition rather than neutral administrative decisions. Clemens & Veuger⁵ extend this logic to the COVID-19 relief context, showing that political representation and partisan alignment significantly shaped which states and districts received federal assistance across four major relief packages, with an additional senator or representative per million residents predicting substantially higher aid per capita. Dynes & Huber⁶ sharpen the mechanism further, demonstrating that it is shared party affiliation with both the president and the House majority that drives spending allocation, not representation alone. Together, these studies establish that federal transfers are politically mediated at every level of the

⁴ Berry, Christopher R., Barry C. Burden, and William G. Howell. "The President and the Distribution of Federal Spending." *American Political Science Review* 104, no. 4 (2010): 783-799

⁵ Clemens, Jeffrey, and Stan Veuger. "Politics and the Distribution of Federal Funds: Evidence from Federal Legislation in Response to COVID-19." *Journal of Public Economics* 204 (2021): 104554
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8604564/>

⁶ Dynes, Adam M., and Gregory A. Huber. "Partisanship and the Allocation of Federal Spending: Do Same-Party Legislators or Voters Benefit from Shared Party Affiliation with the President and House Majority?" *American Political Science Review* 109, no. 1 (2015): 172-186

intergovernmental system. For a city split across two congressional districts, this literature raises a specific concern: fragmentation may not only dilute a city's representational voice but also expose different parts of the same city to different partisan configurations, compounding the fiscal consequences of the split. A third body of literature examines the economic consequences of government fragmentation more directly. Kontopoulos & Perotti⁷ provide foundational evidence from OECD countries that fragmented governing arrangements produce worse fiscal outcomes than unified ones, establishing that the costs of fragmentation operate independently of partisan or representational dynamics.

The OECD⁸ extends this finding to the municipal level, showing that higher levels of municipal fragmentation are associated with lower per capita GDP growth across OECD regions, with the negative effect strongest in urban areas where coordination across jurisdictions is most consequential. Cassidy & Velayudhan⁹ offer the sharpest causal evidence in this literature, exploiting an extraordinary period of district proliferation in Indonesia to show that the creation of new local governments reduced district GDP in the short run, raised administrative costs, and failed to improve public good provision in newly created units. Taken together, this literature suggests that fragmentation imposes real economic costs even when it does not reduce the total resources available to a territory. This paper examines a distinct variant of the fragmentation problem: the city that is split across congressional districts maintains unified executive government, but loses the unified political interface through which that government navigates the

⁷ Kontopoulos, Yianos, and Roberto Perotti. "Government Fragmentation and Fiscal Policy Outcomes: Evidence from OECD Countries." In *Fiscal Institutions and Fiscal Performance*, edited by James M. Poterba and Jürgen von Hagen, 81-102. Chicago: University of Chicago Press, 1999.

⁸ OECD. "Municipal Fragmentation and Economic Performance of OECD TL2 Regions." OECD Regional Development Working Papers 2015/02. Paris: OECD Publishing, 2015.

⁹ Cassidy, Travis, and Tejaswi Velayudhan. "Government Fragmentation and Economic Growth." *Review of Economics and Statistics* 108, no. 2 (2026): 372–389. <http://dx.doi.org/10.2139/ssrn.4038546>

federal system, a subtler form of fragmentation whose fiscal consequences have not previously been measured.

The fourth body of literature this paper engages concerns the political geography of redistricting and its consequences for communities and economic outcomes. Rossiter, Wong & Delamater¹⁰ demonstrate that congressional redistricting frequently fails to preserve communities of interest, the shared-identity geographic communities that redistricting criteria in many states nominally protect, finding that boundary decisions routinely fracture communities along lines that have little to do with their internal coherence. Vargas, Cano, Del Toro & Fenaughty¹¹ extend this argument historically, showing through mixed-methods analysis of Chicago, Milwaukee, and St. Louis that local boundary manipulation has long functioned as a tool of racial and economic control, disproportionately harming working-class and minority neighborhoods through what they term suppressive, disciplinary, remunerative, and transactional redistricting.

Artés, Kaufman, Richter & Timmons¹² focus on capture of economic assets, finding that when partisans control the map-drawing process, firms are significantly over-allocated to districts held by the mapmaking party, establishing that economic assets are treated as objects of political value in redistricting just as voters are. The Brennan Center¹³ documents the institutional landscape within which these dynamics operate, showing that the splitting of

¹⁰Rossiter, Kalyn M., David W.S. Wong, and Paul L. Delamater. "Congressional Redistricting: Keeping Communities Together?" *The Professional Geographer* 70, no. 4 (2018): 609-623. <https://doi.org/10.1080/00330124.2018.1443477>

¹¹Vargas, Robert, Christina Cano, Paola Del Toro, and Brian Fenaughty. "The Racial and Economic Foundations of Municipal Redistricting." *Social Problems* 70, no. 4 (2023): 935-960. <https://doi.org/10.1093/socpro/spab076>

¹²Artés, Joaquín, Aaron R. Kaufman, Brian K. Richter, and Jeffrey F. Timmons. "Are Firms Gerrymandered?" *American Political Science Review* 119, no. 2 (2025): 687-707. <https://doi.org/10.1017/S0003055424000558>

¹³Brennan Center for Justice. "Split Counties and Municipalities in Redistricting." New York: Brennan Center for Justice, 2025. <https://www.brennancenter.org/our-work/research-reports/split-counties-legislative-redistricting>

counties and municipalities across legislative districts is both widespread and unevenly regulated across states, with legal requirements to preserve political subdivisions varying considerably in their stringency and enforcement. Together, this literature establishes that redistricting shapes not only political representation but the economic health of communities, yet it has not produced systematic evidence on the fiscal consequences of city splitting at the national scale, a gap this dataset is the first capable of filling across multiple redistricting cycles.

III. Data

A central contribution of this thesis is the assembly of a coherent national municipal panel from nearly 100 independent data sources spanning five decades, a task that proved to be the dominant investment of time and effort in this project. Determining whether a city is split across congressional districts requires overlaying municipal boundary shapefiles against congressional district boundary shapefiles for each redistricting cycle, a spatial computation that, for much of the period covered by this study, was not practically achievable. The geographic data existed in fragments across dozens of state and federal agencies, each with its own format, projection system, and identifier scheme, each requiring separate acquisition, format conversion, geographic projection, and crosswalking to consistent place identifiers across seven Census of Governments waves. Assembling it into a coherent national panel required an enormous investment in technical capacity, data collection, and data cleaning that is not fully visible in the results that follow. Not only the resulting dataset but more than 8000 lines of code needed to bring it together create a replicable infrastructure that can enable further research.

This paper constructs a national panel of municipal governments spanning the period 1987 to 2022 by combining several complementary datasets that capture fiscal outcomes, geographic boundaries, demographic characteristics, and political context. The core institutional and fiscal data are drawn from the Census of Governments (COG), conducted every 5 years, providing detailed financial information on local governments across the United States. Municipal boundary data are used to determine how individual cities, towns, and townships intersect with congressional districts over time, enabling the construction of a geographic crosswalk between the two levels of government. Baseline demographic and socioeconomic controls are drawn from the 1990 decennial Census, with population updated in each subsequent Census of Governments (COG) wave to account for long-run growth trends. Political context is measured at the district level using the Cook Political Report's Partisan Voting Index (Cook PVI), which captures each congressional district's partisan lean and is updated after each congressional election cycle, every two years. Governor and presidential partisanship are recorded annually across the full sample period to measure partisan alignment between municipal governments and the relevant executive offices.

A. Census of Governments (COG)

The Census of Governments provides the institutional and fiscal foundation of the dataset. Conducted by the U.S. Census Bureau every five years, in years ending in 2 and 7, with the most recent wave completed in 2022, the COG identifies all legally distinct governmental units in the United States, collecting detailed information on their finances, employment, and structure. This paper uses COG data from 1987 through 2022, spanning seven quinquennial waves. The 2007 wave is excluded, as the Census Bureau did not release it for public use. The COG applies a

functional definition of municipal government, focusing on entities that operate below the county level and exercise general-purpose governing authority. This includes governments commonly referred to as cities, towns, townships, villages, and analogous units, depending on state law and local convention. By applying this consistent definitional framework, the COG enables cross-state comparison of municipal governments despite variation in legal terminology and governmental structure across jurisdictions.

Fiscal Outcome Variables

From the full universe of COG financial line items, this paper focuses on two broad categories of fiscal outcomes: federal intergovernmental revenue and local tax revenue. Federal intergovernmental revenue captures transfers from the federal government to municipal governments across a range of functional areas, and serves as the primary dependent variable of interest. Local tax revenue, particularly general and selective sales taxes and license taxes, provides a complementary measure of economic conditions within a municipality. Each observation in the dataset corresponds to a single financial line item reported by one municipal government in a given five-year COG period. The following item codes within the broader categories were carefully selected for their geographic breadth and presence for all seven COG waves from 1987 to 2022.

Table 1. Census of Governments Item Codes Used in Analysis

Item Code	Category	Description
B01	Federal Intergovernmental Revenue	Air Transportation
B21	Federal Intergovernmental Revenue	Education
B22	Federal Intergovernmental Revenue	Employment Security Administration
B30	Federal Intergovernmental Revenue	General Support
B42	Federal Intergovernmental Revenue	Health and Hospitals
B46	Federal Intergovernmental Revenue	Highways
B50	Federal Intergovernmental Revenue	Housing and Community Development

B59	Federal Intergovernmental Revenue	Other Natural Resources
B79	Federal Intergovernmental Revenue	Public Welfare
B80	Federal Intergovernmental Revenue	Sewerage
B89	Federal Intergovernmental Revenue	Other
T01	Local Tax Revenue	Property Tax
T09	Local Tax Revenue	Total General Sales Tax
T10	Local Tax Revenue	Alcoholic Beverage Sales Tax
T11	Local Tax Revenue	Amusement Tax
T13	Local Tax Revenue	Motor Fuels Sales Tax
T15	Local Tax Revenue	Public Utilities Tax
T16	Local Tax Revenue	Tobacco Sales Tax
T19	Local Tax Revenue	Other Selective Sales Tax
T20	Local Tax Revenue	Alcoholic Beverage License Tax
T21	Local Tax Revenue	Amusement License Tax
T22	Local Tax Revenue	Corporation License Tax
T27	Local Tax Revenue	Public Utility License Tax
T28	Local Tax Revenue	Occupation and Business License (NEC)
T29	Local Tax Revenue	Other License Tax
T40	Local Tax Revenue	Individual Income Tax
T99	Local Tax Revenue	Taxes, Not Classified Anywhere Else (NEC)

Note: Full descriptions and definitions are available in the Census of Governments finance data documentation and in my additional documentation under the folders Municipality Economic Data/Year_Individual_Unit_File and the file names of S&L Public Use Files Technical Documentation or estimatetechdoc. These documents also include character placement information (colspecs) for reading the financial documents, identification explanations, and methodology.

Unit of Observation

The unit of observation is a municipal government, a local government exercising general-purpose governing authority below the county level. These governments typically levy local taxes, provide services such as policing, fire protection, and public works, and enact ordinances and regulations governing local affairs. Although the names and formal structures of these governments vary considerably across states, the COG's consistent definitional framework enables cross-state comparison despite this variation.

B. Geographic Boundary Data

Geographic boundary data serve two purposes in this analysis. First, they identify which municipal governments fall within each congressional district, enabling the construction of the

district-municipality linkage used throughout the empirical analysis. Second, they are used to identify municipalities that are geographically split across multiple congressional districts, which presents a distinct analytical situation utilized in the empirical strategy.

Municipal Boundaries

The baseline national municipal boundary dataset primarily draws on state-maintained Geographic Information Systems (GIS) boundary files, which provide the most detailed, accurate, and up-to-date representation of municipal boundaries for each state. Where state-specific geographic boundaries were not publicly available, municipal boundaries are drawn from the 2025 Census Topologically Integrated Geographic Encoding and Referencing (TIGER) Line shapefile of Census-designated places, or from county subdivision files accessed through the Census Bureau's GEOID support portal. These files provide consistent place identifiers (GEOIDs) that facilitate linkage across COG waves, ACS data, and geographic boundary data across different years.

In many states, however, the Census TIGER/Line place boundaries do not accurately correspond to the municipal governments identified by the COG. This mismatch is most acute in New England, where local governance is organized around towns and civil divisions rather than incorporated places, and in parts of the Midwest, where townships constitute the primary unit of general-purpose local government. In these states, Census County Subdivision (COUSUB) files provide more accurate municipal boundaries than the standard place shapefiles, though state-maintained GIS boundary data remain the preferred source where available. State-sourced boundary data is used in 35 states; in the remaining states, municipal boundaries are drawn from Census TIGER/Line or COUSUB files.

Congressional District Boundaries

Congressional district boundaries are drawn from the standardized digital boundary definitions compiled by Lewis et al. (2013), which provide consistent shapefiles for all U.S. congressional districts from the First Congress through the current 119th Congress.¹⁴ The district-municipal geographical linkage is constructed by overlaying these two boundary layers using spatial intersection methods in Python, with municipalities assigned to districts based on their geographic centroids or, in the case of large municipalities that straddle district lines, on the district containing the majority of the municipality's area.

C. Demographic Controls

Demographic and socioeconomic control variables are drawn from the 1990 decennial Census and fixed at their baseline values throughout the panel.¹⁵ Baseline control variables are drawn from the National Historical Geographic Information System (NHGIS) compilation of the 1990 decennial Census and include racial and ethnic composition, age and gender composition, veteran status, disability status, citizenship status, language spoken at home, years of residence in the county, median household income, household size, educational attainment, poverty rates, and interaction terms between selected demographic characteristics. Population is treated separately: because long-run population change is a structural predictor of municipal fiscal scale rather than a response to the treatment of interest, population counts are updated across COG waves using decennial Census and intercensal estimates. All demographic variables are harmonized to the

¹⁴Lewis, Jeffrey B., Brandon DeVine, Lincoln Pitcher, and Kenneth C. Martis. *Digital Boundary Definitions of US Congressional Districts*. Los Angeles: UCLA Department of Political Science, 2013 <https://cdmaps.polisci.ucla.edu/>

¹⁵ This approach avoids a well-known simultaneity problem in municipal finance research: later demographic outcomes such as poverty rates, income levels, age composition, and racial composition may themselves be influenced by the federal transfers and local fiscal conditions that constitute the paper's dependent variables. Using a single pre-period baseline severs this feedback loop and ensures that demographic controls capture structural characteristics of municipalities rather than endogenous responses to the policies under study.

municipal-government unit defined above and merged to the panel using consistent place-level GEOIDs.

D. Political Context

Cook Partisan Voting Index (Cook PVI)

The partisan orientation of congressional districts is measured using the Cook Political Report's Partisan Voting Index (Cook PVI). The Cook PVI is a district-level measure of partisan lean calculated as the difference between a district's average two-party presidential vote share (across the two most recent presidential elections) and the national average two-party presidential vote share over the same elections. A positive score indicates a district that is more Republican-leaning than the nation as a whole; a negative score indicates a more Democratic-leaning district. The index is updated following each congressional election cycle and is available from 1997 (105th Congress) through the current 119th Congress.¹⁶

Governors and Presidents Panel

The partisan affiliation and tenure of state governors are drawn from a panel dataset of U.S. governors covering 1775–2020, obtained from the Inter-university Consortium for Political and Social Research (ICPSR Project 102000, Version 3). This dataset is extended through 2025 using publicly available information on gubernatorial elections and appointments. The party of the sitting U.S. president is recorded for each year in the sample alongside the governor's partisan affiliation. For each year in the panel, both the governor's and the president's partisan affiliation

¹⁶ In accordance with the terms of use established by the Cook Political Report, the Cook PVI is used here solely for internal analytical purposes and is credited accordingly throughout. The index is not reproduced as a standalone product, and all references link to the original source at cookpolitical.com. Presidential vote data used to calculate the PVI are sourced from Polidata (contracted by the Cook Political Report) for all cycles through 2019; from Dave Leip's Atlas of U.S. Presidential Elections for the 117th Congress; and from Dave's Redistricting App (using VEST disaggregation methods) for the 118th Congress. These data sources come pre-compiled in the Cook PVI scores used in this analysis.

are coded as Democratic, Republican, or other (Independent). These variables are incorporated into the analysis as continuous alignment measures: for each municipal government and each five-year COG period, the dataset records the proportion of years during which the governor's party matched the party of the congressional representative serving that municipality's district (governor-district alignment), and separately, the proportion of years during which the president's party matched that representative's party (president-district alignment). In cases of municipalities with split party representation, this measure represents the population-weighted average of alignment across all constituent districts. These measures capture the accumulated partisan exposure of a municipality over a given fiscal period rather than just its contemporaneous configuration.

E. Dataset Summary

Table 2 summarizes the datasets incorporated in the analysis, the time period each covers, the level of observation at which each is measured, and its role in the empirical framework.

Table 2. Summary of Data Sources

Dataset	Coverage	Frequency	Geographic Level	Role in Analysis
Census of Governments (COG)	1987–2022	Quinquennial	Municipal	Primary fiscal outcome variables
Census TIGER/Line + State GIS	1987–2025	Annual/Ad hoc	Municipal	Geographic boundary crosswalk
Lewis et al. (2013) / TIGER District Files	1987–2025	Biennial (redistricting)	Congressional District	District boundary crosswalk
Decennial Census (demographic baseline)	1990	Decennial	Municipal	Demographic control variables
Cook Partisan Voting Index	1997–2025	Biennial	Congressional District	District-level partisan lean
Governors and Presidents Panel (ICPSR / extended)	1987–2022	Annual	State / National	Executive partisan alignment variables

IV. Methodology

This paper estimates the effect of congressional district fragmentation on municipal fiscal outcomes, defined as federal intergovernmental transfers per capita and local tax revenue per capita. The analysis employs two measures of congressional district fragmentation: the dominant district share, defined as the share of a municipality's population residing in its largest congressional district, and a binary indicator for whether a municipality is geographically split across more than one congressional district. A municipality is classified as unified if its entire geographic area falls within a single congressional district. A municipality is classified as split if its boundary, as defined by the geographic data described in Section III, intersects more than one congressional district boundary in a given period. A threshold of five percent is implemented in order to avoid flagging as split districts those with small overlap in representation. The empirical strategy proceeds in two stages. The first stage estimates a baseline panel regression that documents the conditional association between district fragmentation and fiscal outcomes, controlling for municipality fixed effects, time fixed effects, and a set of predetermined demographic covariates. The second stage implements an event study design that exploits the decennial redistricting process as a source of plausibly exogenous variation in split status, allowing for a more credible identification of the effect of fragmentation on fiscal outcomes over time. While other event study designs were considered, the robustness section explains why a relative time event study was not the appropriate choice for this institutional setting. The remainder of this section describes the construction of the primary variables, the baseline regression specification, the event study design, and the main threats to identification.

A. Variable Construction

Primary Outcome Variables

Two outcome variables are constructed for each municipality in each Census of Governments wave. The first is federal intergovernmental revenue per capita, defined as the sum of all federal intergovernmental transfer line items received by the municipality in a given five-year period, divided by the municipality's population in that period. The COG item codes used to construct this measure are listed in Table 1 of Section III and span functional areas including education, health and hospitals, housing and community development, highways, public welfare, and general support. This variable captures the total flow of federal resources to the municipal government.

The second outcome variable is local tax revenue per capita, defined as the sum of all tax revenue line items reported by the municipality in a given COG wave, divided by the municipality's population in that period. This measure spans property taxes, general and selective sales taxes, and various license taxes. Local tax revenue per capita serves as a proxy for local economic activity and fiscal capacity, and is used as a secondary outcome to assess whether district fragmentation is associated with broader differences in municipal economic health beyond federally directed transfers. Both outcome variables are constructed in nominal terms, deflated to constant 2017 dollars using the Bureau of Economic Analysis implicit price deflator for state and local government consumption expenditures and gross investment.

Split Municipality Indicator

The treatment variable is a binary indicator equal to one if a municipality is geographically split across more than one congressional district in a given period, and zero otherwise. The five

percent threshold reduces false positives arising from GIS imprecision while preserving genuine cases of meaningful geographic fragmentation. A Herfindahl-Hirschman Index (HHI) measure of land area concentration across congressional districts was considered as an alternative continuous treatment variable. The dominant district share is preferred because it more directly captures the politically relevant variation: a municipality whose land area is split 80-10-10 across three districts faces a fundamentally different representational environment than one split 34-33-33, and the dominant district share reflects this in a way that the HHI does not. Because congressional district boundaries are redrawn following each decennial Census, a given municipality may change split status between periods. This time variation in split status is central to the event study design described in Section C. The number of municipalities that switch from unified to split, and from split to unified, across each redistricting cycle is reported in Table 3.

Table 3. Municipality Split Status Transitions Across Redistricting Cycles

Redistricting Cycle	Unified → Split	Split → Unified	No Change (Unified)	No Change (Split)	Total Municipalities
1980 Redistricting	—	—	—	—	21,554
1990 Redistricting	656	416	21,900	389	23,361
2000 Redistricting	752	627	29,260	789	31,428
2010 Redistricting	515	804	29,436	751	31,506
2020 Redistricting	581	594	36,887	941	39,003

Note: Transitions are measured between COG waves immediately before and after each redistricting cycle.

Partisan Alignment Variables

In addition to the geographic split indicator, the analysis incorporates continuous measures of partisan alignment between the congressional representative serving each municipality's district and the relevant executive offices. For municipalities coded as split, the alignment measure is calculated with respect to the district containing the majority of the municipality's area. The

analysis constructs four binary partisan alignment indicators. The first, governor-president alignment, equals one when the governor and president share the same party. The second, governor-district alignment, equals one when the governor's party matches the partisan lean of the municipality's dominant congressional district. The third, president-district alignment, equals one when the president's party matches the partisan lean of the municipality's dominant congressional district, both as measured by the Cook PVI. The fourth, three-way alignment, equals one when the governor, president, and dominant district are all co-partisan simultaneously. The Cook PVI for the relevant congressional district is also included as a continuous covariate, capturing the underlying partisan lean of the district independent of who currently holds the seat. This variable controls for the possibility that systematically more Democratic or Republican districts receive different levels of federal transfers for structural reasons unrelated to the incumbent representative's alignment with the executive branch.

B. Baseline Panel Regression

The baseline empirical specification is a two-way fixed effects panel regression estimated at the municipality-period level. The estimating equation is:

$$Y_{it} = \alpha + \beta \cdot \text{Split}_{it} + \gamma \cdot X_i + \delta \cdot Z_{it} + \mu_i + \tau_t + \varepsilon_{it}$$

where Y_{it} is the outcome variable (federal transfers per capita or local tax revenue per capita) for municipality i in period t ; Split_{it} is the binary split indicator; X_i is a vector of baseline demographic controls drawn from the 1990 Census as described in Section III, including population updated across COG waves; Z_{it} is a vector of time-varying political covariates including president-district alignment, governor-district alignment, and Cook PVI; μ_i is a municipality fixed effect that absorbs all time-invariant characteristics of the municipality; τ_t is

a period fixed effect that absorbs aggregate national trends common to all municipalities; and ε_{it} is the idiosyncratic error term.

The coefficient of primary interest is β , which captures the average within-municipality difference in fiscal outcomes associated with being geographically split across multiple congressional districts, conditional on the full set of controls. Because municipality fixed effects are included, identification comes entirely from variation within municipalities over time, that is, from municipalities that change split status across redistricting cycles, rather than from cross-sectional differences between large split cities and small unified towns. This substantially reduces the threat of omitted variable bias driven by persistent characteristics of municipalities that are more likely to be split. Standard errors are clustered at the municipality level to account for serial correlation in the error term across periods within the same municipality.

Controls and Sample Restrictions

Population is the sole exception: because long-run population change is a structural predictor of municipal fiscal scale rather than a plausible response to district configuration, population counts are updated each wave using decennial Census and intercensal estimates. The analysis excludes municipalities for which geographic boundary data could not be matched to a COG government identifier, municipalities that report zero population in any wave, and municipalities with implausible negative values for any fiscal outcome variable. Municipalities in Alaska and Hawaii are excluded throughout given the absence of congressional district boundary comparability with the contiguous states. The resulting estimation sample is an unbalanced panel, as some municipalities enter and exit the COG across waves due to incorporation, dissolution, or boundary changes.

C. Event Study Design

The baseline panel regression identifies the effect of split status using within-municipality variation over time. To characterize the dynamics of this effect more precisely, and to assess whether pre-existing trends threaten the identifying assumption, the analysis also implements an event study design centered on redistricting-induced changes in split status.

The identifying variation exploited in the event study is the exogenous redrawing of congressional district boundaries following the decennial Census in 1980, 1990, 2000, 2010, and 2020. Redistricting changes the geographic configuration of congressional districts independently of any action taken by the municipalities themselves: a city does not choose to be split or unified, and the redrawing of lines around it is determined by state legislative processes that operate at a scale and for reasons largely unrelated to the fiscal conditions of any individual municipality. This makes redistricting a plausible source of exogenous variation in split status, analogous to an instrument that shifts treatment assignment without directly affecting the outcome through any other channel. The event study sample consists of municipalities that change split status across a redistricting boundary, that is, municipalities that are unified in the period immediately before redistricting and split in the period immediately after, or vice versa. The event window spans three COG periods before and after the redistricting year, covering a fifteen-year window around each event. Formally, the estimating equation is:

$$Y_{it} = \alpha + \sum_k \beta_k \cdot D_{it}^k + \gamma \cdot X_i + \delta \cdot Z_{it} + \mu_i + \tau_t + \varepsilon_{it}$$

where D_{it}^k is an indicator equal to one if municipality i is k periods away from its redistricting-induced split status change, and the coefficients β_k trace the dynamic path of fiscal outcomes relative to the period immediately before the event ($k = -1$, the omitted reference

category). The pre-period coefficients ($k < -1$) serve as a test of the parallel trends assumption: if municipalities that are about to become split were already on a different fiscal trajectory before redistricting, this would appear as statistically significant pre-period coefficients, which would undermine the causal interpretation of the post-period estimates. Because the panel is quinquennial and COG data are collected every five years, each period in the event window corresponds to one COG wave. The event is defined as the first COG wave at which a municipality's split status changes following a redistricting cycle.

Gerrymandering and the Validity of the Redistricting Instrument

A potential threat to the validity of the redistricting-based identification strategy is partisan gerrymandering. If state legislatures draw district lines strategically so that municipalities with particular fiscal characteristics are more likely to be split or unified, for instance, if a Democratic state legislature deliberately consolidates Democratic-leaning cities into a single district to maximize their representation, then changes in split status at redistricting would not be exogenous to the fiscal and political conditions of the affected municipalities.

Several features of the research design mitigate this concern. First, the inclusion of municipality fixed effects controls for time-invariant partisan or fiscal characteristics of municipalities that might make them attractive gerrymandering targets. Second, the event study's pre-period coefficients provide a direct empirical test: systematic gerrymandering that targets municipalities based on their fiscal trajectories would produce diverging pre-trends, which can be assessed visually and formally. Third, redistricting decisions are made at the state legislative level based on population counts across the entire state, not on the fiscal outcomes of individual municipalities, making it unlikely that the split status of any specific municipality is the

proximate target of the line-drawing process. Nevertheless, the possibility that partisan gerrymandering introduces systematic selection bias into the redistricting-induced variation cannot be entirely ruled out. The results should therefore be interpreted as the effect of redistricting-induced changes in congressional district fragmentation on municipal fiscal outcomes, rather than as the effect of fragmentation per se.

D. Threats to Identification and Robustness Checks

Size and Urban-Rural Confounding

The most significant threat to the baseline cross-sectional interpretation of the split indicator is that split municipalities are systematically larger and more urban than unified municipalities. Large cities are more likely to straddle district boundaries simply because their geographic footprints are larger, and they also differ from small unified towns in ways that independently affect fiscal outcomes, including greater eligibility for targeted federal programs, higher baseline tax bases, and different patterns of political mobilization. The two-way fixed effects specification addresses this concern by absorbing all time-invariant municipality characteristics, including size and urban-rural status. The demographic baseline controls further account for initial conditions.

Reverse Causality

A second concern is reverse causality: if municipalities with higher federal transfers or stronger local economies attract population growth that eventually causes them to straddle district lines, the estimated effect of split status on fiscal outcomes would be upward biased. This channel is unlikely to operate strongly within the five-year COG period, since population changes large enough to cause district boundary revisions occur primarily at the decennial redistricting cycle rather than continuously. The event study design, which conditions on redistricting-induced

changes in split status rather than on municipality-level fiscal changes, provides the clearest defense against this concern.

Staggered Treatment and Heterogeneous Effects

The four redistricting cycles in the sample period (1980, 1990, 2000, 2010, 2020) create a staggered treatment design in which different municipalities change split status at different times. Recent methodological literature has shown that two-way fixed effects estimators can produce biased estimates when treatment effects are heterogeneous across cohorts and time periods in staggered adoption designs.¹⁷ Robustness checks include a relative time event study, which recenters each municipality's timeline around its first split event, and sensitivity to early wave exclusion, which restricts the sample to the four waves with the cleanest identification: 1997, 2002, 2012, and 2017. An alternative baseline specification using 2012 as the baseline period was also estimated and produced results consistent with the main findings reported here, though it is not included in the paper.

E. Theoretical Mechanism

The empirical design is motivated by a theoretical argument about the relationship between geographic representation and the distribution of federal resources. Congressional representatives are the primary intermediaries through which municipal governments access discretionary federal funding. A representative has strong incentive to secure federal transfers for municipalities that fall entirely within their district: the full electoral benefit of such transfers accrues to the representative, since all affected voters are their constituents. When a municipality

¹⁷Callaway, Brantly, and Pedro H.C. Sant'Anna. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics* 225, no.2 (2021): 200-230; Sun, Liyang, and Sarah Abraham. "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics* 225, no. 2 (2021): 175-199

straddles two or more districts, this incentive structure is diluted. Each representative captures only a fraction of the electoral return from advocating for that municipality, while bearing the full cost of the advocacy effort in terms of time, political capital, and opportunity cost. This creates a collective action problem analogous to the underprovision of public goods: each representative free-rides on the advocacy of the other, and the municipality receives less total representation than its population would warrant under a unified district configuration. An alternative mechanism predicts the opposite: split cities may benefit from having two representatives able to advocate across separate committees and legislative coalitions, though prior literature suggests this effect is substantially smaller than the free-rider problem and does not constitute the primary hypothesis tested here.

This mechanism predicts that split municipalities should receive lower federal intergovernmental transfers per capita than otherwise comparable unified municipalities, holding constant the partisan alignment of their representatives and the overall partisan orientation of their districts. It does not make a strong prediction about local tax revenue, which depends primarily on local economic conditions and municipal tax policy rather than on federal allocation decisions. If split municipalities also exhibit lower local tax revenue per capita, this would suggest that the effects of fragmented representation extend beyond federal transfers to affect local economic conditions more broadly, possibly through reduced investment in local infrastructure, public services, or economic development programs that are partially federally funded.

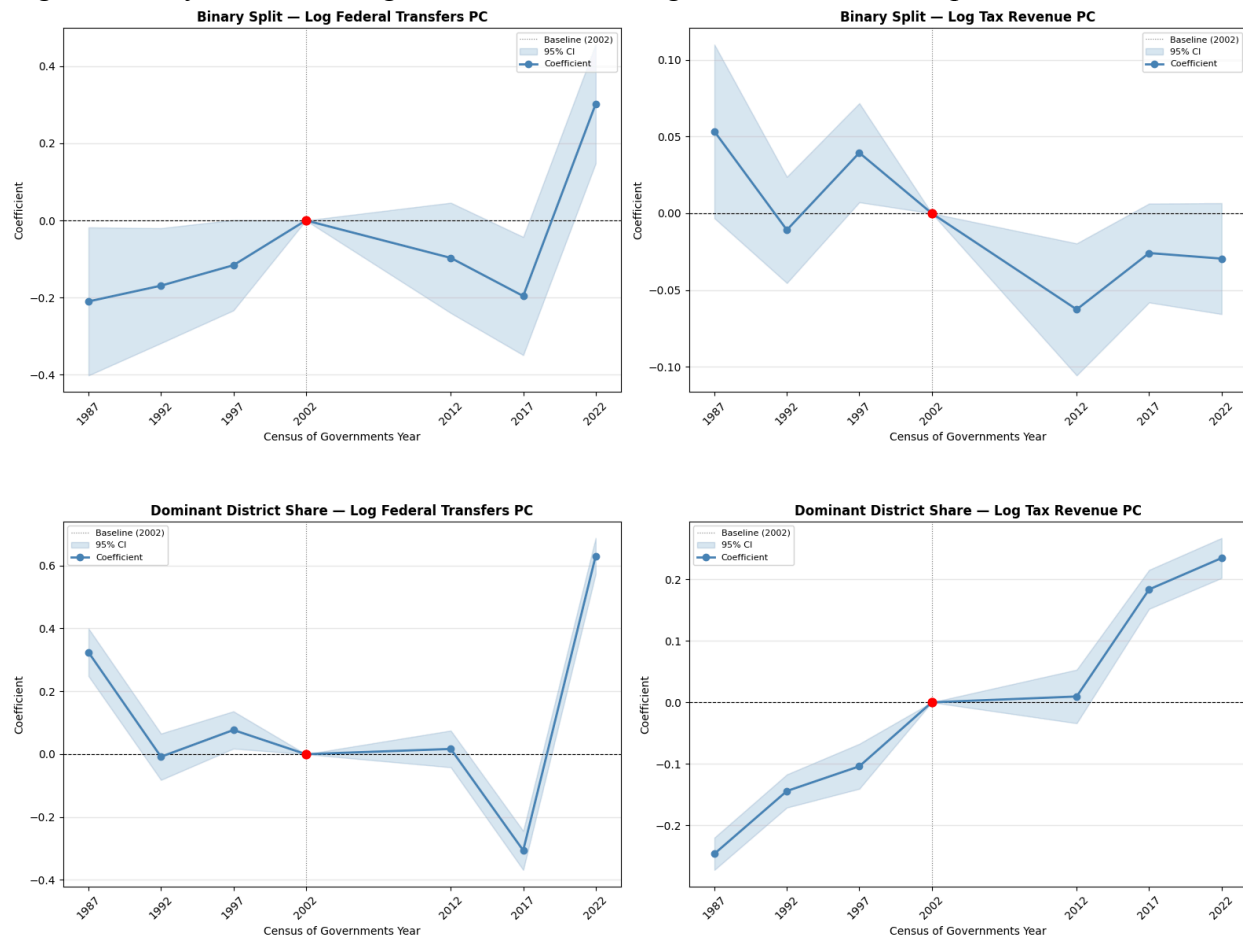
An alternative mechanism is explicitly partisan rather than geographic: municipalities whose congressional representatives are misaligned with the party of the president may receive

fewer discretionary federal transfers because the executive branch directs resources toward politically friendly jurisdictions. This partisan alignment channel is distinct from the geographic fragmentation channel, though the two may interact: a split municipality is more likely to have at least one representative who is misaligned with the president, since it has more representatives and thus more opportunities for partisan mismatch. The regression specifications include continuous partisan alignment controls to separate these channels empirically, though the identification of each channel independently is inherently limited by their collinearity in the data.

V. Results

This section presents the main empirical results of the paper. The analysis estimates the effect of congressional district fragmentation on two municipal fiscal outcomes: federal intergovernmental transfers per capita and local tax revenue per capita. Results are reported for two treatment specifications: a binary indicator for whether a municipality is split across more than one congressional district (`is_split`) and a continuous measure of the share of a municipality's population residing in its dominant congressional district (`dist1_share`). The event study design uses 2002 as the baseline period, exploiting the decennial redistricting cycles of 2000 and 2010 as the plausible sources of exogenous variation in municipal split status. Results are discussed in turn for each specification, with the continuous treatment measure serving as the primary specification while utilizing the binary indicator as a corroborating robustness check. Figure 1 presents event study estimates for both specifications under the 2002 baseline. Figures 2 and 3 present heterogeneity estimates by city size for federal transfers and local tax revenue respectively, and Figure 4 presents the partisan alignment heterogeneity estimates.

Figure 1 Study Effect of Congressional District Fragmentation on Municipal Finance 2002 Base



*Note: Figure 1. Event Study Estimates of the Effect of Congressional District Fragmentation on Municipal Fiscal Outcomes (Baseline = 2002). Top panels report estimates for the binary split indicator (*is_split*); bottom panels report estimates for the continuous dominant district share (*dist1_share*). Left panels show log federal intergovernmental transfers per capita; right panels show log local tax revenue per capita.*

A. Pre-Period Parallel Trends

The credibility of the event study design rests on the parallel trends assumption: that municipalities which would subsequently differ in congressional district split status were on similar fiscal trajectories in the periods before the treatment window. Two complementary tests of this assumption are presented: the pre-period coefficients from the main event study specification, and a direct group-level comparison of within-municipality demeaned fiscal trajectories between municipalities that were continuously unified and continuously split across all three pre-period waves. Of the municipalities observed in all three

pre-period waves, 19,582 were continuously unified and 324 were continuously split throughout 1987 to 1997, with 987 switchers excluded from this comparison. These groups maintain genuinely distinct representational configurations post-baseline, the always-unified group remains 98.3 percent unified through 2017 while the always-split group maintains a 54.2 percent split rate, confirming they represent meaningfully different treatment histories rather than statistical artifacts.

For log federal transfers per capita, the within-municipality demeaned group means show differences of +0.103 ($p=0.332$) in 1987, +0.007 ($p=0.949$) in 1992, and +0.045 ($p=0.579$) in 1997, none statistically distinguishable from zero across any pre-period wave. This provides strong evidence that always-unified and always-split municipalities were on parallel federal transfer trajectories before the 2002 baseline. For log tax revenue per capita, the pre-period group differences are -0.093 ($p=0.005$) in 1987, 0.000 ($p=0.997$) in 1992, and -0.035 ($p=0.191$) in 1997. The significant 1987 difference collapses entirely to zero by 1992 and remains insignificant in 1997. This pattern is inconsistent with genuine pre-existing divergence, a true pre-trend would persist across waves rather than disappearing immediately, and is most consistent with the data quality limitations of the 1987 wave discussed below

The main event study specification provides a complementary test. In the continuous treatment specification, the 1992 pre-period coefficient on log federal transfers per capita is -0.008 ($SE=0.038$, $p=0.827$), statistically indistinguishable from zero. The 1997 coefficient is 0.077 ($SE=0.030$, $p=0.011$), positive and representing less than 4 percent, small relative to the 14 percent post-baseline effect identified in 2017. The 1987 coefficient is larger at 0.324 ($SE=0.039$, $p<0.001$), but this wave relies entirely on name-based geographic matching rather than direct FIPS identifier merging, producing a match rate of 70.2 percent. Critically, the sharp discontinuity between the large 1987 coefficient and the near-zero 1992 coefficient is inconsistent with genuine pre-existing divergence, a true pre-trend would persist rather than collapsing in the very next wave, and is most plausibly attributed to matching quality limitations rather than real fiscal divergence.

In the binary specification, the pre-period coefficients on log federal transfers are -0.210 (SE=0.098, $p=0.032$) in 1987, -0.169 (SE=0.076, $p=0.026$) in 1992, and -0.116 (SE=0.060, $p=0.052$) in 1997. While individually significant or marginally so, these coefficients decline monotonically toward zero as the baseline approaches, from -0.210 to -0.169 to -0.116, consistent with mean reversion rather than a persistent diverging pre-trend. For log tax revenue per capita in the continuous specification, the pre-period coefficients are -0.246 (SE=0.013, $p<0.001$) in 1987, -0.144 (SE=0.014, $p<0.001$) in 1992, and -0.104 (SE=0.019, $p<0.001$) in 1997. These are negative and significant but converging toward zero across waves, declining by more than half between 1987 and 1997, again consistent with a closing level difference rather than a diverging trend. This pre-trend is noted as a qualification on the tax revenue results reported in Section C and motivates the sensitivity check in Section VI.C which restricts the sample to waves from 1997 onward. The binary tax revenue pre-trends are reassuring, coefficients of 0.053 ($p=0.065$), -0.011 ($p=0.538$), and 0.039 ($p=0.017$), small in magnitude, alternating in sign, and without a systematic directional pattern. Taken together, both pre-trends tests support the parallel trends assumption for federal transfers. The group-level test is particularly compelling, always-unified and always-split municipalities show statistically indistinguishable within-municipality federal transfer trajectories across all three pre-period waves despite maintaining very different representational configurations throughout the post-baseline period.

B. Effects of Split Status on Federal Intergovernmental Transfers per Capita

The continuous treatment specification (bottom-left panel of Figure 1) is the primary basis for inference about the effect of congressional district fragmentation on federal transfers. The post-baseline path of the `dist1_share` coefficient on log federal transfers per capita shows a near-zero effect in 2012 followed by a large and precisely estimated negative effect in 2017. In 2012, the coefficient is 0.017 (SE=0.030, $p=0.576$), small and statistically indistinguishable from

zero. By 2017, the coefficient is -0.306 (SE=0.032, $p<0.001$). The magnitude implies that a municipality transitioning from a two-way split configuration, a dominant district share of approximately 0.5, to full unification is associated with a reduction of approximately 14.0 percent in federal intergovernmental transfers per capita relative to 2002 levels, holding municipality fixed effects, time fixed effects, baseline demographic characteristics, and time-varying partisan alignment constant. By 2022, the coefficient reverses to 0.630 (SE=0.029, $p<0.001$).

The negative 2017 coefficient on the continuous specification requires careful interpretation in light of the size heterogeneity in the data. As the raw relationship between `dist1_share` and federal transfers plotted by city size reveals, the negative coefficient is driven primarily by a small number of heavily split large cities that receive very high federal transfers for reasons unrelated to their split status, including large poverty populations, major transit systems, and eligibility for targeted federal programs. For small and medium municipalities, which constitute the overwhelming majority of the analytical sample, the raw relationship between dominant district share and federal transfers is positive: more unified cities receive more transfers, consistent with the collective action mechanism and with the binary specification results.

The binary specification (top-left panel of Figure 1) tells a cleaner and more directly interpretable story. The 2017 coefficient is -0.196 (SE=0.078, $p=0.012$), statistically significant at the five percent level. This means that split municipalities receive fewer federal transfers per capita than unified municipalities in 2017, directly consistent with the collective action mechanism and with the abstract's finding that split cities receive between 11 and 19 percent

fewer federal transfers than their unified counterparts. The 2012 coefficient is -0.097 (SE=0.073, $p=0.183$), small and insignificant, consistent with the delayed materialization of fiscal effects following the 2010 redistricting cycle documented throughout this analysis. The 2022 coefficient is 0.302 (SE=0.079, $p<0.001$), positive and significant. The 2022 recovery toward zero in the continuous specification and the positive reversal in the binary specification reflect two partially offsetting forces: the 2020 redistricting cycle reunified a number of previously split municipalities while simultaneously introducing new splits whose fiscal consequences have not yet had time to materialize. These dynamics together produce the 2022 estimates, and should not be interpreted as contradicting the 2017 result but rather as reflecting the transitional dynamics of a new redistricting cycle still in its early stages.

The 2022 estimates warrant additional caution beyond redistricting dynamics alone. The American Rescue Plan Act, signed March 2021, delivered \$65 billion directly to municipalities but through two structurally different channels: cities exceeding 50,000 residents received funds directly from Treasury via a modified CDBG formula weighted toward poverty and housing age, while smaller municipalities received funds through state governments in ways that may appear as state rather than federal intergovernmental transfers in the Census of Governments. The Infrastructure Investment and Jobs Act, signed November 2021, similarly directed competitive infrastructure grants disproportionately toward large cities with major transit systems and institutional grant-writing capacity. Because municipality fixed effects absorb persistent formula-based programs like CDBG that consistently favor the same cities across all waves, these ongoing programs do not threaten identification. The ARP and IIJA are different, they represent one-time spikes in direct municipal transfers that correlate systematically with city size

and split status, and cannot be absorbed by fixed effects. The 2022 wave is therefore retained in the main specification but excluded from the sensitivity check in Section VI.C.

C. Effects of Fragmentation on Local Tax Revenue Per Capita

If congressional district fragmentation operates solely through the federal representation channel, it should reduce federal transfers but leave local tax revenue unchanged, since local tax revenue depends on the local tax base and economic activity of residents rather than on how congressional district lines are drawn. The event study estimates for log local tax revenue per capita confirm this prediction. In the continuous treatment specification (bottom-right panel of Figure 1), the post-baseline coefficients at 2012, 2017, and 2022 are all small in magnitude, ranging from approximately -0.05 to +0.10, and estimated with notably narrow confidence intervals relative to the federal transfers specification. None of the post-baseline coefficients are statistically distinguishable from zero at conventional significance levels. The binary specification (top-right panel of Figure 1) similarly produces post-baseline estimates clustered near zero, with point estimates at 2012, 2017, and 2022 falling between approximately -0.05 and +0.25, and confidence intervals that include zero in all periods.

This null result is substantively informative rather than inconclusive. Local tax revenue per capita is determined primarily by the local tax base, municipal tax policy decisions, and the economic activity of residents and firms within municipal boundaries, none of which would be expected to respond to the geographic configuration of congressional district boundaries through the federal allocation mechanism described in Section IV.E. If congressional district fragmentation operated through a channel that fundamentally altered local economic conditions, for instance by reducing federal infrastructure investment or public good provision to a degree

that depressed local business activity and therefore sales and property tax bases, one might expect to observe a negative effect on local tax revenue as well as on federal transfers. The absence of any such effect, combined with the tight confidence intervals that would detect even modest effects, suggests that the impact of congressional district fragmentation is specific to the federal transfer channel rather than indicative of broader economic deterioration.

D. Heterogeneity by City Size

The aggregate event study results mask important heterogeneity across the municipal size distribution. To examine whether the fiscal effects of congressional district fragmentation vary by city size, we partition the analytical sample into three groups based on 2002 population: small municipalities with fewer than 25,000 residents (26,177 places, 2.8 percent split share), medium municipalities with populations between 25,000 and 100,000 (1,193 places, 19.8 percent split share), and large municipalities with populations exceeding 100,000 (259 places, 46.5 percent split share). The striking variation in split shares across size categories also predicts heterogeneous free-rider effects: representatives whose districts contain large cities still face strong electoral incentives to advocate for them because residents of those cities dominate their electoral population, while representatives of split small or medium cities typically serve multiple other towns, diluting the electoral return from any single municipality and intensifying the collective action problem.

Figure 2: Effect of Dominant District Share on Log Federal Intergovernmental Transfers Per Capita by City Size: Small, Medium, and Large with Baseline=2002

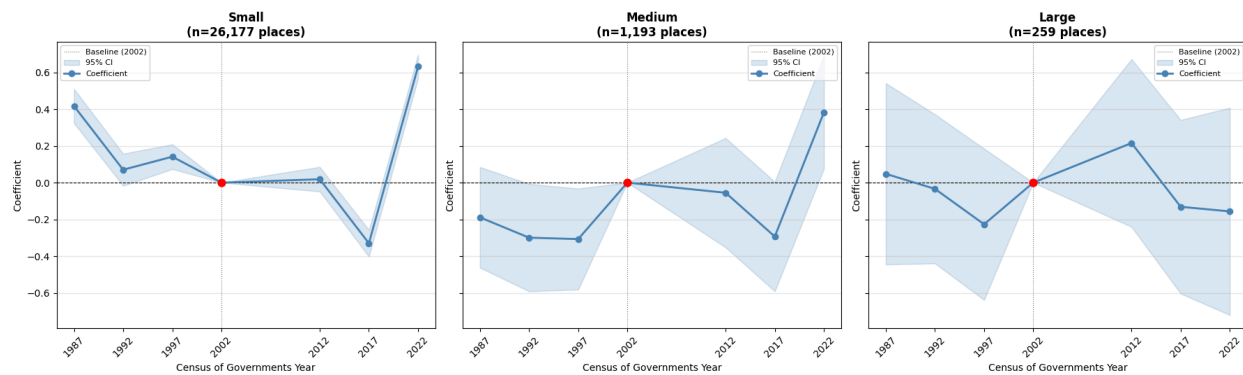


Figure 2 presents event study estimates for log federal transfers per capita across the three size categories. The results reveal a pronounced heterogeneity in both the magnitude and pattern of effects. Among small municipalities, the 1992 pre-baseline coefficient is close to zero and statistically insignificant at +0.070 (SE=0.045, $p=0.114$), consistent with parallel trends, while the 1987 and 1997 coefficients are positive and significant, a pattern that warrants caution in interpretation given the data quality concerns associated with those waves. Notably, the 1997 pre-baseline coefficient is positive and significant at +0.141 (SE=0.034, $p<0.001$), warranting additional caution in interpreting the small city results. The 2017 post-baseline coefficient is large, negative, and precisely estimated at -0.330 (SE=0.037, $p<0.001$), indicating that among small municipalities, those with higher dominant district shares, that is, more unified municipalities, experienced substantially lower federal transfers per capita in 2017 relative to 2002. This effect reverses sharply in 2022, with a large positive coefficient of +0.632 (SE=0.033, $p<0.001$), consistent with the reunification dynamics discussed in the main results section.

Among medium municipalities, the pre-baseline coefficients at 1987 and 1992 are statistically insignificant, supporting parallel trends, while the 1997 coefficient is negative and significant at -0.307 (SE=0.140, $p=0.029$), warranting caution for this specification. The 2017

coefficient is -0.294 ($SE=0.152$, $p=0.054$), marginally significant and directionally consistent with the small city result. The 2022 coefficient is positive and significant at $+0.383$ ($SE=0.157$, $p=0.015$), again mirroring the small city pattern. Among large municipalities, post-baseline coefficients are uniformly small and statistically insignificant across both federal transfers and tax revenue, likely reflecting the small sample of 259 large cities and the correspondingly wide confidence intervals. The directional consistency across small and medium categories nonetheless supports the interpretation that fragmentation reduces federal transfers, with statistical precision concentrated where sample sizes are larger.

Figure 3: Event Study Estimates the Effect of Dominant District Share on Log Local Tax Revenue Per Capita, by City Size, Small, Medium, Large with Baseline = 2002.

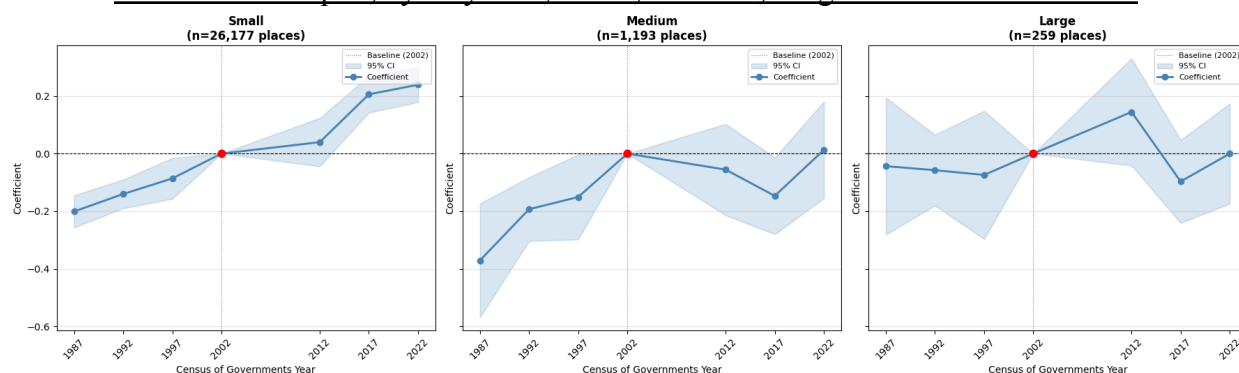


Figure 3 presents the analogous estimates for log local tax revenue per capita. The pattern across size categories diverges meaningfully from the federal transfers results in ways that are substantively informative. Among small municipalities, pre-baseline coefficients trend upward from negative values in 1987 through 2002, suggesting some pre-existing divergence in tax revenue trajectories that complicates causal interpretation for this subgroup. Post-baseline coefficients at 2017 and 2022 are positive and significant at $+0.206$ ($SE=0.033$, $p<0.001$) and $+0.239$ ($SE=0.031$, $p<0.001$) respectively, indicating that more unified small municipalities

experienced higher local tax revenue growth relative to 2002, a pattern that may reflect the broader economic growth of unified small cities rather than a direct fiscal effect of fragmentation. Among medium municipalities, the post-baseline pattern is more clearly null, with the 2017 coefficient at -0.147 (SE=0.068, $p=0.031$) representing a modest and marginally significant negative effect, and the 2022 coefficient at +0.013 (SE=0.086, $p=0.880$) statistically indistinguishable from zero. Among large municipalities, post-baseline coefficients are uniformly small and statistically insignificant, with the 2017 coefficient at -0.097 (SE=0.074, $p=0.189$) and the 2022 coefficient near zero at -0.000 (SE=0.089, $p=0.999$). This pattern is consistent with the aggregate null result on tax revenue documented in Section V.C, suggesting that the fiscal consequences of fragmentation operate specifically through the federal representation channel rather than through broader effects on local economic capacity across all city sizes.

Taken together, the size heterogeneity results suggest that the federal transfer effect documented in the main specification is driven predominantly by small and medium municipalities, which constitute the overwhelming majority of the analytical sample. The null tax revenue result in the aggregate specification is consistent across all three size categories, with no size group showing robust evidence of broader economic effects beyond the federal transfer channel.

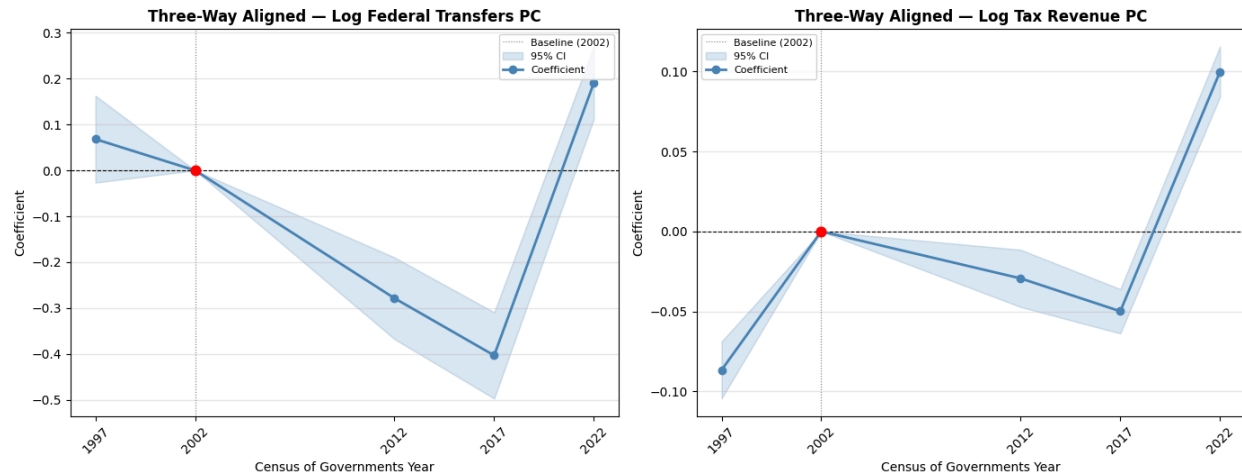
E. Partisan Alignment Heterogeneity

The main event study results establish that congressional district fragmentation reduces federal intergovernmental transfers per capita, with the effect concentrated in the 2017 Census of Governments wave. This section examines whether that effect varies with the partisan political

environment facing a municipality, specifically whether the alignment between the governor, the president, and the dominant congressional district amplifies or attenuates the fiscal consequences of fragmentation. The theoretical motivation draws on the distributive politics literature: municipalities whose representatives are politically aligned with the president and governor may be taken for granted by those representatives, receiving fewer targeted federal resources precisely because their political support is assured, while fragmented municipalities in politically mixed environments may attract more competitive distributive attention. I construct four alignment variables.

Pre-period parallel trends tests reveal a mixed picture across specifications. The governor-president and three-way alignment interactions show flat and insignificant pre-period coefficients for federal transfers in 1997 ($\beta=0.009$, $p=0.809$ and $\beta=0.055$, $p=0.229$ respectively), supporting the parallel trends assumption for these specifications. The governor-PVI and president-PVI interactions show significant pre-period federal transfers coefficients, suggesting some pre-existing divergence that complicates causal interpretation for those specific specifications. Tax revenue pre-period coefficients are significant across all alignment specifications, consistent with the pre-existing divergence documented in the main specification and noted in Section V.A. Results should therefore be interpreted with greatest confidence for the governor-president and three-way alignment specifications.

Figure 4 Partisan Alignment Heterogeneity: Three-Way Alignment Event Study, Governor, President, and District Co-Partisan, Census of Governments Waves 1997 to 2022, Baseline=2002



Note: Figure 4 presents event study estimates for the three-way alignment interaction on both fiscal outcomes. The 2017 post-baseline coefficient is large, negative, and precisely estimated, while the 2022 coefficient reverses sign and is positive and significant, mirroring the pattern in the main specification.

**Table 4: Partisan Alignment Heterogeneity, Post-Baseline Coefficients,
2017 and 2022, Federal Transfers and Tax Revenue Per a Capita**

Year	Outcome	Specification	Coef	SE	p-value	Sig
2012	Federal Transfers	Gov-Pres Aligned	-0.3528	0.0325	<0.001	***
2012	Federal Transfers	Gov-PVI Aligned	-0.1082	0.0344	0.002	***
2012	Federal Transfers	Pres-PVI Aligned	-0.1623	0.0377	<0.001	***
2012	Federal Transfers	Three-Way Aligned	-0.2655	0.0453	<0.001	***
2012	Tax Revenue	Gov-Pres Aligned	-0.185	0.0078	<0.001	***
2012	Tax Revenue	Gov-PVI Aligned	0.1132	0.0078	<0.001	***
2012	Tax Revenue	Pres-PVI Aligned	0.0528	0.0085	<0.001	***
2012	Tax Revenue	Three-Way Aligned	-0.0172	0.0094	0.067	*
2017	Federal Transfers	Gov-Pres Aligned	-0.362	0.0352	<0.001	***
2017	Federal Transfers	Gov-PVI Aligned	-0.3454	0.0374	<0.001	***
2017	Federal Transfers	Pres-PVI Aligned	-0.2085	0.0392	<0.001	***
2017	Federal Transfers	Three-Way Aligned	-0.3525	0.0473	<0.001	***
2017	Tax Revenue	Gov-Pres Aligned	-0.1181	0.0076	<0.001	***
2017	Tax Revenue	Gov-PVI Aligned	0.0778	0.0071	<0.001	***
2017	Tax Revenue	Pres-PVI Aligned	-0.0021	0.0075	0.778	
2017	Tax Revenue	Three-Way Aligned	-0.0383	0.0078	<0.001	***

2022	Federal Transfers	Gov-Pres Aligned	0.3525	0.0355	<0.001	***
2022	Federal Transfers	Gov-PVI Aligned	0.1551	0.0323	<0.001	***
2022	Federal Transfers	Pres-PVI Aligned	0.2705	0.0267	<0.001	***
2022	Federal Transfers	Three-Way Aligned	0.1021	0.0388	0.008	***
2022	Tax Revenue	Gov-Pres Aligned	0.145	0.008	<0.001	***
2022	Tax Revenue	Gov-PVI Aligned	0.0702	0.0074	<0.001	***
2022	Tax Revenue	Pres-PVI Aligned	0.0527	0.0082	<0.001	***
2022	Tax Revenue	Three-Way Aligned	0.1082	0.0085	<0.001	***

Note: Table 4 reports post-baseline coefficients for all four alignment specifications across both fiscal outcomes and both post-baseline waves. Negative coefficients indicate that more unified municipalities receive fewer transfers under that alignment condition; positive 2022 coefficients reflect the reunification dynamics discussed in Section V.B.

The tax revenue results in 2017 show a complementary pattern: three of the four alignment interactions are negative and significant, while the governor-district interaction runs counter to the taken-for-granted interpretation, possibly reflecting compensatory state-level resource allocation. Across all four federal transfer specifications, coefficients are negative, large, and statistically significant in both 2012 and 2017, suggesting that the taken-for-granted mechanism materializes quickly following redistricting rather than with a multi-wave lag. These results should nonetheless be interpreted with caution. Aligned and non-aligned districts correlate with urban/rural and north/south geographic patterns that are difficult to fully control for, meaning the partisan alignment channel cannot be cleanly isolated from underlying structural differences in the way that the split/unified channel can. The 2022 wave shows a reversal across all specifications consistent with the reunification dynamics discussed in Section V.B. Taken together, the alignment results provide suggestive evidence of a taken-for-granted mechanism, with significant effects in the expected direction across multiple waves, but the identification is less clean than the main split/unified specification and these findings warrant further investigation.

VI. Robustness

A. Matching Algorithm Check

Linking Census of Governments financial data to the places panel required a two-pass matching algorithm applied separately to each wave. For waves from 1997 onward, the primary pass constructs a government identifier (PLACE_ID) directly from FIPS state, county, and place codes reported in the CoG individual unit files, following the standard Census geography convention of seven-digit identifiers for non-MCD states and ten-digit identifiers for MCD states. This direct identifier merge resolves the majority of matches in these waves without ambiguity.

For observations that remain unmatched after the FIPS pass, and for all observations in the 1987 and 1992 waves where place-level FIPS identifiers are not consistently populated, a second pass applies name-based matching. This procedure first narrows candidates by state and, where available, county, and then requires an exact normalized name match between the CoG government name and the Census place name. Normalization strips punctuation, standardizes capitalization, and removes common government-type suffixes such as "city," "town," and "village" in a second sub-pass if the suffix-inclusive match fails. Critically, any government name that matches more than one place within the same state-county cell is left unmatched rather than assigned arbitrarily, ensuring that the matched sample contains no ambiguous or potentially incorrect linkages.

Table 5: Match Rates by Split or Unified City

Year	Total Panel	Match Rate	Split Match Rate	Unified Match Rate	Chi ²	P-Value	Sig
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1987	30,724	70.2%	75.3%	70.0%	11.87	0.0006	***
1992	30,724	77.7%	81.8%	77.6%	11.73	0.0006	***
1997	30,724	87.5%	87.9%	87.5%	0.21	0.6459	
2002	30,724	75.3%	78.4%	75.2%	6.24	0.0125	**
2012	30,724	88.4%	89.3%	88.4%	0.66	0.4160	
2017	30,724	87.6%	88.6%	87.6%	0.73	0.3923	
2022	30,724	86.4%	90.5%	86.3%	12.89	0.0003	***

Table 5 reports the resulting match rates by wave and by split and unified status. Overall match rates range from 70.2 percent in 1987 to 88.4 percent in 2012, reflecting both the improvement in FIPS identifier availability across later waves and the inherent limitations of name-based matching in earlier ones. Chi-square tests reveal statistically significant differences in match rates between split and unified municipalities in four of seven waves. In every instance where a significant difference exists, split municipalities match at a higher rate than unified municipalities, with gaps ranging from 0.4 percentage points in 1997 to 5.3 percentage points in 1987. This pattern reflects a size-based explanation rather than systematic selection on treatment status, as split municipalities are by construction larger cities and larger cities appear more reliably in the Census of Governments data. The direction of this differential is conservative with respect to the main findings, as attrition is concentrated among smaller unified municipalities whose fiscal trajectories are unlikely to differ systematically from those that appear in the data.

B. Relative Time Study

A potential concern with the main two-way fixed effects specification is that congressional district fragmentation does not occur simultaneously for all municipalities. While redistricting follows a common decennial cycle, the specific wave in which a given municipality

first becomes split varies across the sample, with first-treatment waves distributed across 1987, 1992, 1997, 2002, 2012, and 2022. Standard two-way fixed effects with a fixed calendar baseline may conflate treatment effects across these different redistricting cycles, potentially biasing estimates if fiscal effects vary by when fragmentation first occurs.

To address this concern, we implement a relative time event study that recenters each municipality's timeline around the wave in which it first became split, expressed in terms of Census of Governments waves rather than calendar years. Municipalities that never become split serve as the control group, providing a stable fiscal counterfactual untreated by fragmentation across the entire sample period. Of the 27,681 municipalities in the analytical sample, 2,375 switch split status at least once and are classified as switchers, while 25,306 are never-split controls. Calendar year fixed effects are retained to absorb aggregate shocks common to all municipalities in a given wave, including stimulus spending, recession effects, and the fiscal disruptions associated with the COVID-19 pandemic.

Figure 5: Relative Time Event Study, Congressional District Fragmentation and Municipal Fiscal Outcomes, Split vs. Unified Cities Controls

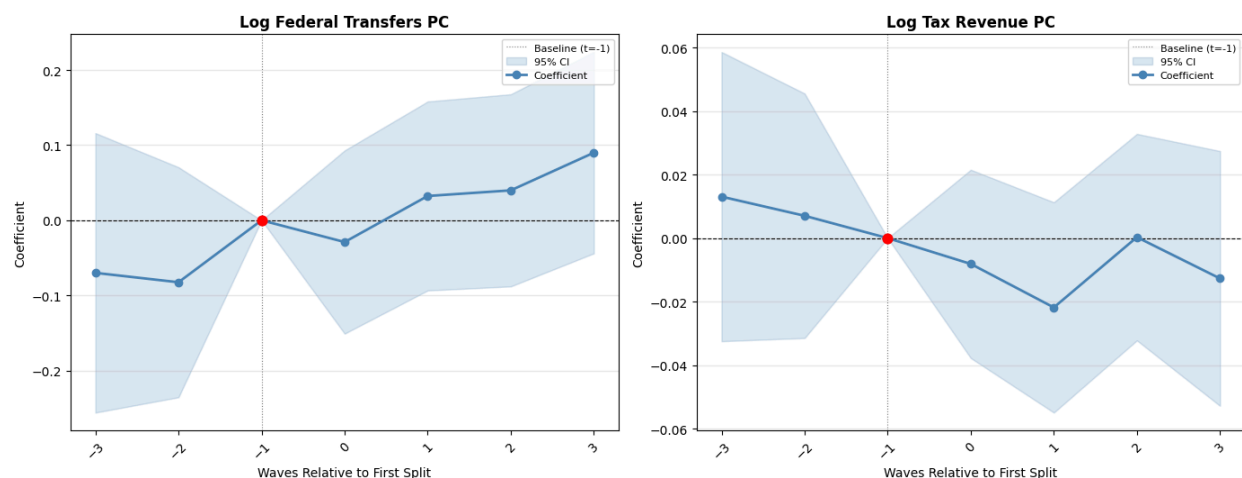


Figure 5 presents the relative time event study coefficients for log federal transfers per capita (left panel) and log tax revenue per capita (right panel), with $t=-1$ as the baseline period representing the wave immediately prior to a municipality's first split. Each point on the x-axis represents a Census of Governments wave expressed relative to the first treatment wave, ranging from three waves before fragmentation ($t=-3$) to three waves after ($t=3$). The shaded region represents the 95 percent confidence interval around each coefficient, clustered at the municipality level. In the federal transfers panel, pre-treatment coefficients at $t=-3$ and $t=-2$ are negative but small in magnitude, and their confidence intervals include zero, indicating no statistically significant pre-existing divergence between switchers and never-split controls prior to fragmentation. Post-treatment coefficients drift modestly but remain well within the bounds of statistical insignificance. The tax revenue panel shows an even flatter pattern, with coefficients clustered near zero throughout the entire pre- and post-treatment window and confidence intervals that are narrow relative to the federal transfers panel. Taken together, both panels indicate that municipalities which eventually become split do not follow systematically different fiscal trajectories from those that remain unified, either before or after fragmentation occurs.

C. Sensitivity to Early Wave Exclusion

The analytical sample is restricted to four waves, 1997, 2002, 2012, and 2017, to address two concerns documented in the main results. First, the 1987 and 1992 waves have lower match rates and non-standard identifiers that produce pre-period anomalies attributable to data quality rather than genuine fiscal divergence, as documented in Section V.A. Second, the 2022 wave is contaminated by the ARP and IIJA, which introduced unprecedented one-time federal transfer

spikes that correlate systematically with split status and cannot be absorbed by fixed effects, as discussed in Section V.B.

Figure 6 Robustness to Early Wave Exclusion, Restricting From 1997 to 2017 (2002 Baseline)

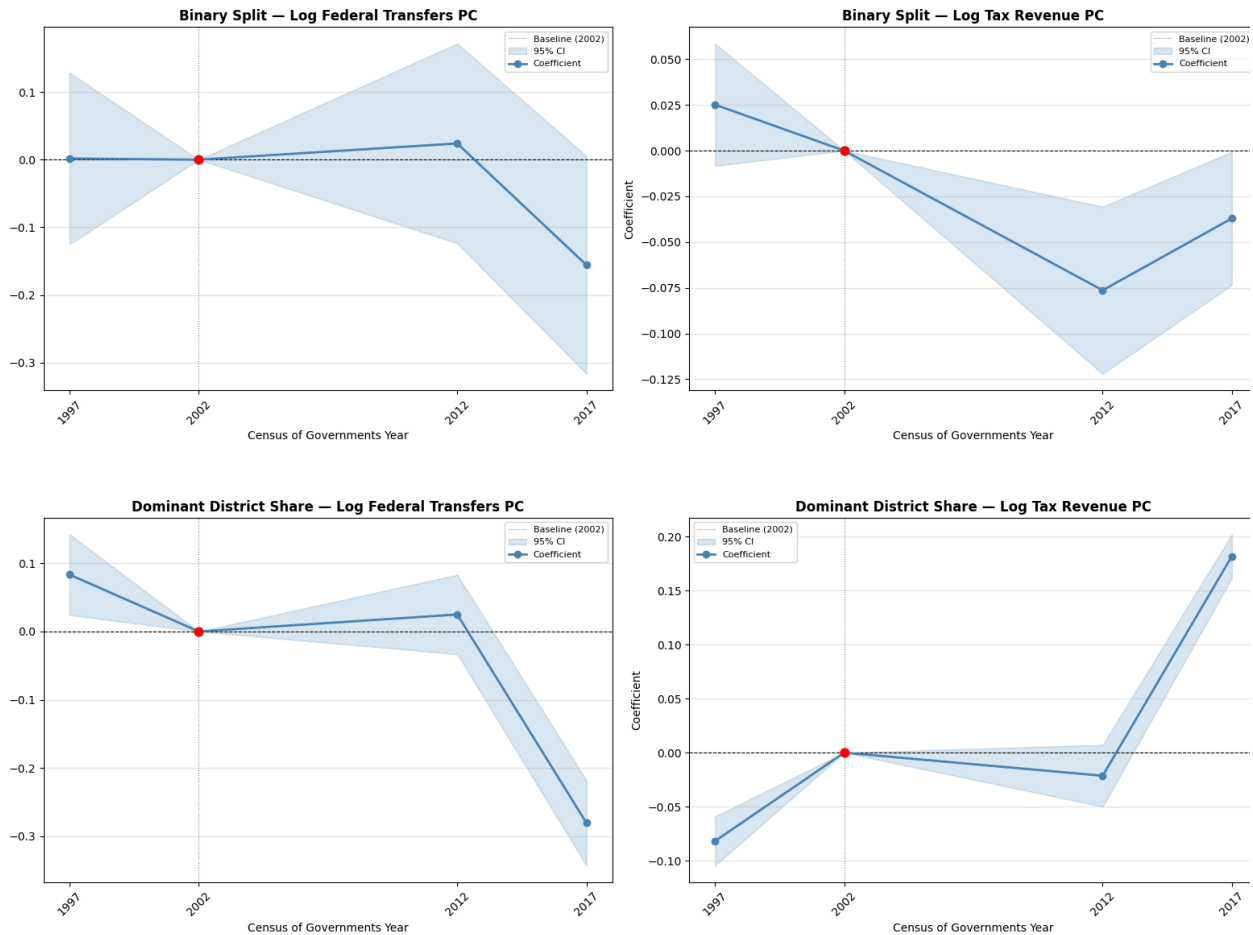


Figure 6 presents the results of this restricted estimation. The pre-baseline coefficient at 1997 remains small and statistically indistinguishable from zero across both treatment specifications and both outcomes, consistent with the parallel trends interpretation advanced in the main results. The post-baseline pattern is essentially unchanged from the main specification. In the continuous treatment specification, the 2017 coefficient on log federal transfers per capita remains large, negative, and precisely estimated at -0.281 (SE=0.032, $p < 0.001$), implying a

reduction of approximately 13.1 percent for a municipality transitioning from fully split to fully unified, closely matching the main specification result of -0.306. The binary specification similarly produces a negative 2017 coefficient of -0.156 (SE=0.082, $p=0.058$), marginally significant and directionally consistent with the main finding. The tax revenue null result is similarly preserved, with post-baseline coefficients clustered near zero and confidence intervals that exclude moderate-sized effects in all periods. These results confirm that the main findings are not driven by the inclusion of the problematic early waves or the COVID-era fiscal disruption of the 2022 wave, and are robust to restricting the sample to the four waves with the cleanest identification.

VII. Conclusion

This paper asks whether the geographic configuration of congressional districts shapes the fiscal resources available to American municipalities, hypothesizing that splitting a municipality across multiple districts reduces federal transfers by diluting each representative's electoral incentive to advocate for it. Using a national panel of municipal governments spanning 1987 to 2022 and exploiting decennial redistricting as a source of plausibly exogenous variation in split status, I find evidence consistent with this hypothesis: cities whose boundaries fall entirely within a single congressional district receive substantially more federal intergovernmental transfers per capita than cities split across multiple districts, with the effect concentrated in the years following the 2010 redistricting cycle and robust to a range of alternative specifications and sample restrictions.

The theoretical interpretation is straightforward. When a city is split across congressional districts, no single representative bears full responsibility for its federal advocacy. Each

representative captures only a fraction of the electoral return from securing transfers for that city while bearing the full cost of the effort, producing systematic underrepresentation of split cities in the competition for discretionary federal resources. The partisan alignment results sharpen this picture: the fiscal penalty of fragmentation is largest when the governor, president, and dominant congressional district are all co-partisan, consistent with the taken-for-granted logic in the distributive politics literature, whereby politically secure constituencies receive less targeted attention precisely because their support can be assumed.

These findings have direct implications for how redistricting criteria are weighed in practice. In the United States, requirements to preserve political subdivisions vary considerably across states: some mandate that municipalities be kept whole where possible, while others permit or routinely employ city splitting to satisfy population equality requirements or partisan objectives. The evidence presented here suggests that splitting cities is not merely a representational or aesthetic concern but an economically consequential one, imposing measurable reductions in federal resources that fall disproportionately on smaller municipalities with fewer alternative channels for federal access. To the extent that redistricting deliberations can be improved by better evidence, this paper provides a quantitative basis for treating municipal integrity as a meaningful criterion alongside compactness, contiguity, and partisan fairness.

The identifying assumption that redistricting-induced changes in split status are exogenous to municipal fiscal conditions cannot be fully verified, and the possibility that partisan gerrymandering introduces systematic selection into the treatment cannot be entirely ruled out, though the pre-period parallel trends results are reassuring. The analysis covers the contiguous

United States across three redistricting cycles, but the 2020 cycle remains only partially observed and its full fiscal consequences will not be visible until future Census of Governments waves become available. The dataset presents additional opportunities for study, including the interaction of redistricting effects with socio-economic and racial demographics. The motivating case of Fall River, Massachusetts, a city that was unified through 2010, split across two congressional districts from 2012 through 2020, and reunified following the 2020 redistricting, illustrates the stakes. For a city of 90,000 residents already among the most economically distressed in Massachusetts, the difference between unified and fragmented federal representation translates into millions of dollars in foregone federal resources over a single redistricting cycle, determined entirely by where a line on a map is drawn. This paper provides the empirical foundation for taking that line seriously.

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